Question 1

(a) Using Chebyshev's Theorem

Chebyshev's theorem states that for any dataset, at least $1 - \frac{1}{k^2}$ of the data lies within k standard deviations of the mean. To find the range where at least 75% of the data lies:

1. Set $1 - \frac{1}{k^2} = 0.75$.
2. Solve for k:
$$1 - \frac{1}{k^2} = 0.75 \implies \frac{1}{k^2} = 0.25 \implies k^2 = 4 \implies k = 2.$$
3. The range is:
$$\text{Mean} \pm k \cdot \text{Standard Deviation} = 20 \pm 2 \cdot 4 = [12, 28].$$

Answer: The range is $[12, 28]$.

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(b) Response Times Analysis

(i) Mean Absolute Deviation (MAD) and Standard Deviation (SD) for Dataset X

Dataset $X = \{12, 17, 21, 26\}$:

1. **Mean** (μ):
$$\mu = \frac{12 + 17 + 21 + 26}{4} = \frac{76}{4} = 19.$$
2. **MAD**:
$$\text{MAD} = \frac{|12 - 19| + |17 - 19| + |21 - 19| + |26 - 19|}{4} = \frac{7 + 2 + 2 + 7}{4} = \frac{18}{4} = 4.5.$$
3. **SD**:
$$\text{Variance} = \frac{(12 - 19)^2 + (17 - 19)^2 + (21 - 19)^2 + (26 - 19)^2}{4} = \frac{49 + 4 + 4 + 49}{4} = \frac{106}{4} = 26.5.$$
$$\text{SD} = \sqrt{26.5} \approx 5.15.$$

Since $4.5 \leq 5.15$, MAD $\leq$ SD.

(ii) Correlation Coefficient Between X and Y

Dataset $Y = \{14, 19, 23, 27\}$:

1. **Mean of Y (μ_Y):**

$$\mu_Y = \frac{14 + 19 + 23 + 27}{4} = \frac{83}{4} = 20.75.$$

2. **Covariance ($\mathbf{Cov}(X, Y)$):**

$$\text{Cov}(X, Y) = \frac{(-7)(-6.75) + (-2)(-1.75) + (2)(2.25) + (7)(6.25)}{4} = \frac{47.25 + 3.5 + 4.5 + 43.75}{4} = \frac{99}{4}.$$

3. **Correlation Coefficient (r):**

$$r = \frac{\text{Cov}(X, Y)}{\text{SD}_X \cdot \text{SD}_Y} = \frac{24.75}{5.15 \cdot 4.82} \approx 0.99.$$

Interpretation: The correlation coefficient $r \approx 0.93$ indicates a strong positive linear relationship between the response times of the two optimization techniques.

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(c) Network Transmission Times Analysis

(i) Five-Number Summary

Data: $\{25, 30, 42, 55, 60, 120\}$:

1. **Minimum:** 25.
2. Q_1**:** The median of the lower half $\{25, 30, 42\}$ is 30.
3. **Median:** The median of the entire dataset is $\frac{42+55}{2} = 48.5$.
4. Q_3**:** The median of the upper half $\{55, 60, 120\}$ is 60.
5. **Maximum:** 120.

(ii) Interquartile Range (IQR)

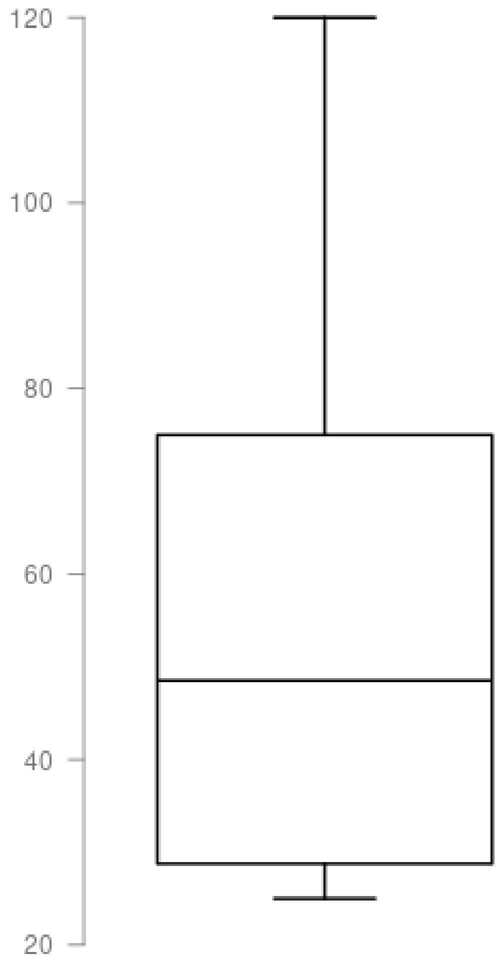
$$\text{IQR} = Q_3 - Q_1 = 60 - 30 = 30.$$

(iii) Outliers Using $1.5\times$ IQR Rule

1. Lower bound: $Q_1 - 1.5 \times \text{IQR} = 30 - 45 = -15$.
2. Upper bound: $Q_3 + 1.5 \times \text{IQR} = 60 + 45 = 105$.
3. **Outliers:** Any data point outside $[-15, 105]$. Here, 120 is an outlier.

(iv) Box-and-Whisker Plot

The plot show the five-number summary with a whisker extending to 105 and an outlier at 120.



(v) Skewness

The Box-and-Whisker Plot has a longer upper whisker, indicating the data is **right-skewed**.

(vi) Effect of Adding 130 ms

The new maximum becomes 130, and the upper whisker extends further, making the skewness more pronounced.

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Question 2

(a) Number of Possibilities

The number of ways to choose 5 coffee, 4 tea, and 3 juice from 12 people is:

$$\binom{12}{5, 4, 3} = \frac{12!}{5!4!3!} = 27720.$$

(b) Selecting Test Cases

1. Total ways to choose 4 test cases from 7: $\binom{7}{4} = 35$.
2. Subtract the cases where none of the first 3 are selected: $\binom{4}{4} = 1$.
3. **Total valid ways:** $35 - 1 = 34$.

(c) Coefficient of $x^9y^5z^6w^3$ in $(x + 2y + 3z + w)^{23}$

The coefficient is:

$$\binom{23}{9, 5, 6, 3} \cdot 2^5 \cdot 3^6 = \frac{23!}{9!5!6!3!} \cdot 32 \cdot 729.$$

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Question 3

(a) σ-Algebra Conditions

A set $\mathcal{F}$ is a σ-algebra if:

1. $\Omega \in \mathcal{F}$.
2. If $A \in \mathcal{F}$, then $A^c \in \mathcal{F}$.
3. If $A_1, A_2, \dots \in \mathcal{F}$, then $\bigcup_{i=1}^{\infty} A_i \in \mathcal{F}$.

(b) Independence of A^c and B^c

If A and B are independent:

$P(A^c \cap B^c) = 1 - P(A \cup B) = 1 - [P(A) + P(B) - P(A \cap B)] = 1 - P(A) - P(B) + P(A)P(B).$

$$P(A^c)P(B^c) = 1 - P(A) - P(B)(1 - P(A)) = (1 - P(A))(1 - P(B)).$$

Thus, $P(A^c \cap B^c) = P(A^c)P(B^c)$, so A^c and B^c are independent.

(c) Biased Coin Toss

Sample space: $\{HH, HT, TH, TT\}$.

1. $P(A) = P(\text{first toss is heads}) = \frac{1}{3}$.
2. $P(B) = P(\text{second toss is heads}) = \frac{1}{3}$.
3. $P(A \cap B) = P(HH) = \frac{1}{9}$.

Since $P(A \cap B) = P(A)P(B)$, A and B are independent.

(d) Generalized Bayes' Theorem

Using the definition of conditional probability and the Law of Total Probability:

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)} = \frac{P(B \mid A)P(A)}{\sum_{i=1}^{n} P(B \mid H_i)P(H_i)}.$$

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Question 4

(a) Disease Probability

Using Bayes' theorem:

$$P(\text{Disease} \mid \text{Positive}) = \frac{P(\text{Positive} \mid \text{Disease})P(\text{Disease})}{P(\text{Positive})}.$$

$P(\text{Positive}) = P(\text{Positive} \mid \text{Disease})P(\text{Disease}) + P(\text{Positive} \mid \text{No Disease})P(\text{No Disease})$.

$$P(\text{Positive}) = 0.9 \cdot 0.01 + 0.05 \cdot 0.99 = 0.0585.$$

$$P(\text{Disease} \mid \text{Positive}) = \frac{0.9 \cdot 0.01}{0.0585} \approx 0.1538.$$

(b) Defective Product Probability

Using Bayes' theorem:

$$P(\text{Machine B} \mid \text{Defective}) = \frac{P(\text{Defective} \mid \text{Machine B})P(\text{Machine B})}{P(\text{Defective})}.$$

$P(\text{Defective}) = P(\text{Defective} \mid \text{Machine A})P(\text{Machine A}) + P(\text{Defective} \mid \text{Machine B})P(\text{Machine B})$.

$$P(\text{Defective}) = 0.05 \cdot 0.6 + 0.1 \cdot 0.4 = 0.07.$$

$$P(\text{Machine B} \mid \text{Defective}) = \frac{0.1 \cdot 0.4}{0.07} \approx 0.5714.$$

(c) Probability of Owning a Dog Given a Cat

Using Bayes' theorem:

$$P(\text{Dog} \mid \text{Cat}) = \frac{P(\text{Cat} \mid \text{Dog})P(\text{Dog})}{P(\text{Cat})}.$$

$$P(\text{Cat} \mid \text{Dog}) = 0.22, \quad P(\text{Dog}) = 0.36, \quad P(\text{Cat}) = 0.30.$$

$$P(\text{Dog} \mid \text{Cat}) = \frac{0.22 \cdot 0.36}{0.30} = 0.264.$$

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Question 5

(a) Probability Distribution of X

1. **Value of c:**

$$3c + 3c + 6c = 1 \implies 12c = 1 \implies c = \frac{1}{12}.$$

2. **Cumulative Probability for $X < 1$:**

$$P(X < 1) = P(X = -1) + P(X = 0) = 3c + 3c = 6c = \frac{6}{12} = 0.5.$$

(b) Probability Distribution of Defective Bulbs

Let X be the number of defective bulbs in 3 selected bulbs:

$$P(X = k) = \frac{\binom{2}{k}\binom{6}{3-k}}{\binom{8}{3}}, \quad k = 0, 1, 2.$$

$$P(X = 0) = \frac{\binom{2}{0}\binom{6}{3}}{\binom{8}{3}} = \frac{20}{56}, \quad P(X = 1) = \frac{\binom{2}{1}\binom{6}{2}}{\binom{8}{3}} = \frac{30}{56}, \quad P(X = 2) = \frac{\binom{2}{2}\binom{6}{1}}{\binom{8}{3}} = \frac{6}{56}.$$